

Benchmarking Service — Developer Guide

Last modified: 2026-09-03T03:37:35Z

Hand-maintained, unlike the generated `results/12run-*.md` files which stamp themselves. Bump the line above when you edit this guide.

A task-oriented guide to driving the Benchmarking Service over its RESTful API. Every `curl` below is derived from the flows we exercised end-to-end (gsm8k single-turn, tau2 multi-turn) on the `ykt3` and `kind-rossocctl` clusters.

- Reference the machine-readable contract in `openapi.json` / `openapi.yaml`.
- Design rationale (what the Service can and cannot enact) lives in `SERVICE_DESIGN_DECISIONS.md` and `KUBECTL_DEPENDENCY_INVENTORY.md`.

In a hurry? §2 gets you a token, §6.0 is one benchmark run start to finish as copy-pasteable `curl`, and §6.1 is the same thing as a single command.

Contents

- 1. Mental model (read this first)
 - Endpoint map
- 2. Getting a caller token
- 3. Onboarding a benchmark (one-time, per cluster/instance)
 - 3.1 Instance config file (`instances/<encoded-iss-host>.json`)
 - 3.2 Infrastructure resources (baked into the benchmark definitions)
 - 3.3 Workload secrets (provisioned out-of-band as cluster Secrets)
 - 3.4 MLflow + OTEL collector (optional, for reports)
- 4. Instance-specific Service config (`/config`)
- 5. Benchmark lifecycle
 - 5.0 Discover what's available
 - 5.1 Deploy (create the MCP tool + A2A agent)
 - 5.2 Wait until Ready
 - 5.3 Submit a run
 - 5.4 Poll run status / list runs
 - 5.5 Get results (report)
 - 5.6 Download output files from S3
 - 5.7 Tear down
- 6. End-to-end examples (validated flows)
 - What you need on the client side
 - 6.0 Run #1 start to finish, on the `ykt3` Service driving `ykt2` workloads
 - 6.1 The same thing in one command (`benchmarking-cli`)
 - 6.2 Other validated flows
 - 6.3 The whole 12-run matrix in one command (`reference/run-12.py`)
- 7. Known limits & error codes
- 8. Extending the catalog (adding or changing a benchmark)
 - 8.1 What a definition holds
 - 8.2 Add a new benchmark
 - 8.3 Change an existing benchmark

- [8.4 What stays runtime-mutable \(for contrast\)](#)

1. Mental model (read this first)

The Service is **pure-Python and HTTP-only**. It never calls a cluster API (no kubectl, no kubeconfig). It talks to the Rossoctl/kagenti backend over HTTPS and to Keycloak for tokens. Two consequences shape everything below:

- 1. **Two independent tokens.**
 - The **caller JWT** you present in **Authorization: Bearer ...** is used *only* for attribution (**preferred_username**) and routing (**iss** selects which instance config applies). It is **never forwarded** upstream.
 - For each request the Service mints its **own** ROPC token (the instance's **benchmarker** credential, password grant) and presents *that* to Rossoctl.
- 2. **The Service enacts only what HTTP allows.** It deploys workloads (agents/tools), runs benchmarks, reads reports, and exports to S3. It **cannot** create cluster Secrets or overlay per-agent ConfigMaps — those are provisioned out-of-band, and the Service reports on them (see §3 onboarding and the [424](#) / [422](#) responses).

Endpoint map

Group	Endpoints	Auth
Identity	GET /hello , GET /healthz (public, unschematized)	caller JWT
Config	GET /config , PUT /config	benchmarker only
Discovery	GET /namespaces , GET /benchmarks , GET /benchmarks/{name}	caller JWT
Raw workloads	GET/POST /agents , GET/DELETE /agents/{ns}/{name} , same for /tools	caller JWT
Benchmark deploy	POST /benchmarks/{name}/deploy , DELETE /benchmarks/{name}/deploy , GET /benchmarks/{name}/status	caller JWT
Run lifecycle	POST /benchmarks/{name}/runs , GET .../runs , GET .../runs/{run_id}	caller JWT
Reporting	GET .../runs/{run_id}/report , GET /benchmarks/{name}/report	caller JWT

2. Getting a caller token

The caller JWT is a normal Keycloak token from the instance's realm. Obtain it via the password grant (the same Direct-Access-Grants flow the dev/test users use). The realm and token endpoint are derived from the instance **iss** ([<iss-origin>/realms/<realm>/protocol/openid-connect/token](#)).

```
# --- Environment (pick the block for your target cluster) ---

# Option A – ykt3 (OpenShift; realm/client "kagenti"), the cluster the e2e runs
were validated on:
export SVC="https://benchmarking.apps.ykt3.example.com"      # Benchmarking
Service base URL
export KC="https://keycloak.apps.ykt3.example.com"          # Keycloak base
(iss origin)
export REALM="kagenti"                                     # realm from the
iss
export KC_CLIENT="kagenti"                                  # public client
with Direct Access Grants

# Option B – kind-rossoctl (local; realm/client "rossoctl"):
#   The instance's iss is http://keycloak.localtest.me:8080/realms/rossoctl,
but in-cluster the
#   Service reaches Keycloak via keycloak_backchannel_url. You still fetch YOUR
token from the
#   externally reachable route below (localtest.me resolves to 127.0.0.1).
# export SVC="http://benchmarking.localtest.me:8080"
# export KC="http://keycloak.localtest.me:8080"
# export REALM="rossoctl"
# export KC_CLIENT="rossoctl"

# Attribution matters: config endpoints require preferred_username ==
"benchmarker".
# Any valid realm user works for deploy/run/report.
# Never hardcode the password – supply it out-of-band (env, secret manager, or
an
# interactive prompt). e.g.: read -rs -p "benchmarker password: " KC_PASS;
export KC_PASS
export KC_USER="benchmarker"
export KC_PASS="${KC_PASS:?set KC_PASS in your environment; do not commit it}"

export TOKEN=$(curl -s "$KC/realms/$REALM/protocol/openid-connect/token" \
  -d grant_type=password \
  -d client_id="$KC_CLIENT" \
  -d username="$KC_USER" \
  -d password="$KC_PASS" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["access_token"])')

# Sanity check: the Service echoes the validated claims + which instance you
routed to.
curl -s "$SVC/hello" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

GET /hello returns {iss, preferred_username, rossoctl_base_url, claims}. If you get 401, the token is missing/invalid; 403 issuer not in scope means no instance file matches the token's iss (see §3.1).

Why the kind block differs: in-cluster the public iss host is unreachable, so the instance file sets keycloak_backchannel_url and the Service dials that for JWKS/ROPC. This only affects the Service's own dialing — your caller token still comes from the externally reachable Keycloak route (Option B above).

3. Onboarding a benchmark (one-time, per cluster/instance)

These steps are **out-of-band** — the Service verifies them but cannot perform them.

3.1 Instance config file (`instances/<encoded-iss-host>.json`)

One file per instance keys the whole thing to the `iss`. Loaded at startup from `settings.instances_dir`. Shape:

```
{
  "iss": "https://keycloak.apps.ykt3.example.com/realms/kagenti",
  "keycloak_backchannel_url": null,
  "rossoctl_base_url": "https://kagenti-backend.kagenti-
system.svc.cluster.local:8000",
  "service_credential": {
    "client_id": "kagenti",
    "client_secret": "",
    "username": "benchmarker",
    "password": "<benchmarker-password>"
  },
  "mlflow": { "tracking_url": null },
  "s3": { "bucket": null },
  "mcp_endpoint_template": null,
  "agent_endpoint_template": null,
  "workload_otel": null
}
```

- `service_credential` is the ROPC identity the Service presents to Rossoctl (must exist in the realm with `firstName/lastName` set, or Keycloak 400s with "Account is not fully set up").
- `mcp_endpoint_template` / `agent_endpoint_template` are only needed when the workloads live on a **different** cluster reachable via external routes (use `{service}/{namespace}` placeholders). Leave `null` for co-located in-cluster workloads.
- `mlflow` / `s3` can be seeded here or set later via `PUT /config` (§4).

3.2 Infrastructure resources (baked into the benchmark definitions)

The Service sends CPU/memory requests+limits in the create body, so no post-create patch is needed. Current values (from `benchmarks/registry.py`):

Workload	requests	limits
MCP tool	500m CPU / 512Mi	4 CPU / 4Gi
A2A agent	500m CPU / 512Mi	4 CPU / 2Gi

3.3 Workload secrets (provisioned out-of-band as cluster Secrets)

The Service has **no Secrets API**. It references these by name in the deploy body; if a Secret is missing the workload never becomes Ready, and the run precheck returns **424** naming exactly what to provision.

Benchmark	Secret (name → key)	Consumed by	Purpose
gsm8k	hf-secret → hf-token	tool	HuggingFace dataset access
gsm8k / tau2 / appworld	openai-secret → apikey	agent (+ tau2/appworld tool)	LiteLLM API key

Provision them on the workload cluster before deploying, e.g.:

```
kubectl -n team1 create secret generic hf-secret --from-literal=hf-token="$HF_TOKEN"
kubectl -n team1 create secret generic openai-secret --from-literal=apikey="$LITELLM_KEY"
```

openai-secret must exist before you run the Service — provision it out-of-band. On clusters where the `rossoctl` platform chart owns the agent namespaces (e.g. `team1`), the chart only manages `openai-secret` when its Helm value `secrets.openaiApiKey` is set. Leave that value **empty** and create the Secret yourself (command above / a secret manager / External Secrets), so the credential lives outside Helm values and a later `helm upgrade` **cannot** overwrite it with an empty key. When the value is empty the chart skips the Secret entirely and its install NOTES print a **WARNING: secrets.openaiApiKey is NOT set** reminder. If you instead let the chart template the key, every `helm upgrade` re-applies whatever is in the release values — an empty value there silently zeroes the key mid-flight and every LLM call fails until the Secret is restored and the workload pods are **rollout restarted** (pods read `apikey` via `secretKeyRef` only at startup).

3.4 MLflow + OTEL collector (optional, for reports)

Reporting is fail-soft: if MLflow client-creds aren't configured, runs still succeed and export to S3, but `report.ndjson` is empty and the report endpoints return **409**. To enable, provision an OTEL collector (forwards agent spans to MLflow) out-of-band and set the MLflow read creds via `PUT /config`.

4. Instance-specific Service config (`/config`)

`GET/PUT /config` are **benchmarker-only** (`preferred_username == "benchmarker"`; else **403**). They set only what the Service itself enacts — MLflow (read side) and S3. Attempting to set a workload credential is rejected with **422** (`extra="forbid"`). Secrets are redacted in responses.

```
# View effective config for your instance (file defaults + any overrides),
# secrets redacted.
curl -s "$SVC/config" -H "Authorization: Bearer $TOKEN" | python3 -m json.tool

# Point the Service's result sink at an S3 bucket + wire up MLflow read creds.
curl -s -X PUT "$SVC/config" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{
    "s3": {
      "bucket": "rossoctl-benchmarking",
      "region": "us-east-1",
      "access_key_id": "AKIA...REDACTED",
```

```

        "secret_access_key": "REDACTED",
        "public_read": false
    },
    "mlflow": {
        "tracking_url": "https://mlflow.apps.ykt3.example.com",
        "client_id": "mlflow-client",
        "client_secret": "REDACTED",
        "token_url":
"https://keycloak.apps.ykt3.example.com/realms/kagenti/protocol/openid-
connect/token",
        "insecure_tls": false
    }
}'

```

Rejected example (workload cred → 422):

```

curl -s -o /dev/null -w '%{http_code}\n' -X PUT "$SVC/config" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{"hf-secret": "..."}' # -> 422

```

5. Benchmark lifecycle

The happy path is: **deploy** → **wait for Ready** → **run** → **poll** → **report** → **download from S3**.

Every block below is copy-pasteable once these four variables are exported (see §2 for how to get the token; §6.1 wraps this whole section in a single command if you would rather not paste):

```

export SVC="https://benchmarking-rossoctl-system.apps.ykt3.hcp.res.ibm.com" #
Service base URL
export BENCH=gsm8k # gsm8k | tau2 | appworld
export SCOPE="namespace=team1&agent=tool_calling&experiment=default"
export TOKEN="..." # from §2; never echo this
# Self-signed OpenShift route? add -k to every curl, or: export CURL_OPTS=-k

```

5.0 Discover what's available

```

curl -s "$SVC/benchmarks" -H "Authorization: Bearer $TOKEN" | python3 -m
json.tool
curl -s "$SVC/benchmarks/gsm8k" -H "Authorization: Bearer $TOKEN" | python3 -m
json.tool
curl -s "$SVC/namespaces" -H "Authorization: Bearer $TOKEN" | python3 -m
json.tool

```

Three benchmarks are registered: **gsm8k** (single-turn), **tau2** (multi-turn, runs a server-side user-simulator LLM), **appworld** (registered; see §7).

5.1 Deploy (create the MCP tool + A2A agent)

`POST /benchmarks/{name}/deploy` creates both the shared MCP tool (`exgentic-mcp-<name>`) and the agent (`exgentic-a2a-<agent>-<name>[-<experiment>]`). Returns `201`.

```
curl -s -X POST "$SVC/benchmarks/gsm8k/deploy" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{
  "agent": "tool_calling",
  "namespace": "team1",
  "experiment": "default"
}' | python3 -m json.tool
```

Request fields (`DeployBenchmarkRequest`):

Field	Default	Meaning
agent	tool_calling	agent flavor (only <code>tool_calling</code> today)
model	null	deploy-time model, baked into agent env; <code>null</code> → benchmark default (<code>openai/Qwen3.6-35B-A3B</code>)
namespace	team1	target namespace
experiment	default	non-default suffixes the agent name so variants coexist
authbridge_enabled	false	inject the AuthBridge sidecar with the cluster-default pipeline (layer-2 knob)
plugin_preset	null	layer-3 preset (<code>auth-only ibac-only full</code>); forwarded to the backend as <code>pluginPreset</code> → <code>AgentRuntime.spec</code> → operator renders the per-agent pipeline. Requires <code>authbridge_enabled=true</code>
plugins	null	per-plugin policy overrides as <code>["NAME:POLICY"]</code> tokens (<code>POLICY = enforce observe off</code>); forwarded as <code>plugins</code> . Requires <code>authbridge_enabled=true</code>
on_error	null	chain-default policy (<code>enforce observe off</code>); forwarded as <code>onError</code> . Requires <code>authbridge_enabled=true</code>
plugin_config_file	null	rejected with 422 — a local filesystem path with no HTTP analog; use <code>plugin_preset/plugins/on_error</code> instead

Model swap (run #4 pattern): the deploy endpoint always (re)creates the *shared* tool, so re-deploying with a new model 409s on the existing tool. Deploy the model-swapped agent alone via `POST /agents` under a distinct experiment instead:

```
# Generate the agent body with build_agent_request semantics, then POST /agents directly.
curl -s -X POST "$SVC/agents" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{
  "name": "exgentic-a2a-tool-calling-gsm8k-azmini",
  "namespace": "team1",
```

```
        "container_image": "ghcr.io/exgentic/exgentic-a2a-tool_calling",
        "image_tag": "latest",
        "env_vars": [
            {"name": "MCP_URL", "value": "http://exgentic-mcp-gsm8k-
mcp.team1.svc.cluster.local:8000/mcp"},
            {"name": "LLM_MODEL", "value": "openai/Azure/gpt-4o-mini"},
            {"name": "EXGENTIC_SET_AGENT_MODEL", "value": "openai/Azure/gpt-4o-
mini"},
            {"name": "OPENAI_API_KEY", "value_from": {"secret_key_ref": {"name":
"openai-secret", "key": "apikey"}}}
        ],
        "service_ports": [{"name": "http", "port": 8080, "target_port": 8000}]
    }' | python3 -m json.tool
```

5.2 Wait until Ready

```
curl -s "$SVC/benchmarks/gsm8k/status?
namespace=team1&agent=tool_calling&experiment=default" \
-H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

Returns `tool_ready` / `agent_ready` booleans plus raw `readyStatus`. Poll until both are `true`. If a workload stays not-Ready, the run precheck (§5.3) will name the missing Secret.

5.3 Submit a run

`POST /benchmarks/{name}/runs` runs a **cluster-API-free precheck** (tool+agent deployed and Ready) then returns `202` with a `run_id`. Prechecks:

- `409` — not deployed (`POST .../deploy` first).
- `424` — deployed but not Ready; the message names required Secret(s) (e.g. `hf-secret`).

```
export RUN=$(curl -s -X POST "$SVC/benchmarks/gsm8k/runs" \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
-d '{
    "agent": "tool_calling",
    "namespace": "team1",
    "experiment": "default",
    "max_tasks": 10,
    "max_parallel_sessions": 1,
    "timeout_seconds": 600
}' | python3 -c 'import sys,json; print(json.load(sys.stdin)["run_id"])')
echo "run_id=$RUN"
```

Run fields (`RunRequest`) are all **run-time** knobs:

Field	Default	Notes
<code>max_tasks</code>	<code>1</code>	number of benchmark tasks to evaluate
<code>max_parallel_sessions</code>	<code>1</code>	concurrency

Field	Default	Notes
<code>timeout_seconds</code>	<code>300</code>	whole-run wall-clock ceiling; raise for large tau2 runs
<code>agent / namespace / experiment</code>	—	must match a deployed agent
<code>model</code>	<code>null</code>	not forwarded per-session; model is fixed at deploy time

Timeout tuning (learned e2e): tau2 with `max_tasks=20 max_parallel_sessions=4` exceeded the 900s default and failed; re-running with `timeout_seconds=1800` succeeded (~915s).

5.4 Poll run status / list runs

```
# One run's full state (status, summary, per-task results, artifacts once
exported).
curl -s "$SVC/benchmarks/gsm8k/runs/$RUN" -H "Authorization: Bearer $TOKEN" |
python3 -m json.tool

# All runs for this benchmark scoped to your instance.
curl -s "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" |
python3 -m json.tool
```

`status` moves `pending` → `running` → `succeeded|failed`. On success, `summary` carries `{total, succeeded, evaluated_pass, pass_rate, wall_seconds}`. A simple poll loop:

```
# Terminal statuses are succeeded | failed | error | cancelled – a loop that
waits only for
# succeeded/failed spins forever on the other two.
while ;; do
    S=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization:
Bearer $TOKEN" \
        | python3 -c 'import sys,json; print(json.load(sys.stdin)["status"])')
    echo "status=$S"
    case "$S" in succeeded|failed|error|cancelled) break ;; esac
    sleep 10
done

# The one-line verdict.
curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer
$TOKEN" > /tmp/run.json
python3 - <<'EOF'
import json
d = json.load(open("/tmp/run.json")); s = d["summary"]
print("%s pass_rate=%s %s/%s wall=%.0fs" % (
    d["status"], s["pass_rate"], s["evaluated_pass"], s["total"],
    s["wall_seconds"]))
EOF
```

5.5 Get results (report)

Two report views, both reading structured records from MLflow (return `409` if MLflow isn't configured for the instance — see §3.4):

```
# Per-run report: records filtered to this run's session ids, plus its S3
artifacts.
curl -s "$SVC/benchmarks/gsm8k/runs/$RUN/report" -H "Authorization: Bearer
$TOKEN" | python3 -m json.tool

# Experiment rollup: time-windowed aggregates across an experiment's runs.
curl -s "$SVC/benchmarks/gsm8k/report?experiment=default&window_h=3" \
-H "Authorization: Bearer $TOKEN" | python3 -m json.tool
```

Per-run report is `{run_id, benchmark, experiment, trace_count, records[], artifacts[]}`; each record (`MLflowTraceRecord`) has per-session timing breakdown, LLM/tool latencies + token counts, infra CPU/mem, and the evaluation outcome.

5.6 Download output files from S3

When the instance has an S3 `bucket` set, each completed run is exported (fail-soft) and its objects appear in the run state under `artifacts[]` (and in the per-run report). Layout:

```
<prefix>/<preferred_username>/<encoded-iss>/<benchmark>/<run_id>/
run.json                # the run summary (always written)
report.ndjson           # one record per task (empty if MLflow
unconfigured)
report.parquet          # analytics-friendly (only when there are records)
token_report.ndjson     # lean per-task token view, projected from the
records
token_report.parquet
span_report.ndjson      # one row per OTEL span per task, incl. the span
name
span_report.parquet
manifest.json           # self-describing index of the above (never lists
itself)
```

`span_report.*` is the evidence layer under the other two: `report/token_report` give per-task *counts*, `span_report` gives the spans those counts were derived from, so "did this task really make one model call, or did we lose a span?" is answerable from the artifacts alone. Its columns are a fixed whitelist (`mlflow_report.SPAN_ROW_KEYS`) — span attributes can carry prompts, and these objects are public-read, so nothing outside that list is published.

The S3 URL is fully determined — you can construct it

Objects are addressed as `<url-root>/<key>`, and the key is the layout above. For the bucket these runs use:

URL root	<code>https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/</code>
Key	<code><s3.prefix>/<preferred_username>/<encoded-iss-host>/<benchmark>/<run_id>/<artifact></code>

So a real, working URL — anonymously readable, no credentials, no AWS CLI:

```
https://rossoctl-benchmarking.s3.us-east-1.amazonaws.com/ykt3-to-
ykt2/benchmark/keycloak-keycloak.apps.ykt2.hcp.res.ibm.com-realms-
rossoctl/gsm8k/20260902215628-79e0713a/report.ndjson
```

`s3.prefix` is per instance (`ykt3-to-ykt2/` on the OpenShift instance, `kind/` on KinD), and the issuer host is scheme-stripped with `./://` replaced by `-`. **The bucket is public and anonymously listable**, so treat everything exported as published.

```
# The authoritative list – name, format, size and the full URL, straight from
the run state.
curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization: Bearer
$TOKEN" > /tmp/run.json
python3 - <<'EOF'
import json
for a in json.load(open("/tmp/run.json"))["artifacts"]:
    print("%-8s %9d %s" % (a["format"], a["size_bytes"], a["url"]))
EOF

# Download every artifact by URL – plain curl, no AWS credentials needed.
PREFIX=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H
"Authorization: Bearer $TOKEN" \
| python3 -c 'import sys,json; print(json.load(sys.stdin)
["artifacts_prefix"])')
mkdir -p "/tmp/benchmarking/$PREFIX" && cd "/tmp/benchmarking/$PREFIX"
for f in run.json report.ndjson report.parquet token_report.ndjson
token_report.parquet \
    span_report.ndjson span_report.parquet manifest.json; do
    curl -sS -O "https://rossoctl-benchmarking.s3.us-east-
1.amazonaws.com/$PREFIX/$f"
done && ls -la

# Or discover the set from the manifest, which indexes every object but itself.
curl -s "https://rossoctl-benchmarking.s3.us-east-
1.amazonaws.com/$PREFIX/manifest.json" \
| python3 -m json.tool

# Private bucket instead of public_read? Then use the AWS CLI with credentials.
aws s3 cp "s3://rossoctl-benchmarking/$PREFIX/" ./ --recursive
```

5.7 Tear down

```
curl -s -X DELETE "$SVC/benchmarks/gsm8k/deploy?
namespace=team1&agent=tool_calling&experiment=default" \
-H "Authorization: Bearer $TOKEN" -o /dev/null -w '%{http_code}\n' # -> 204
```

Deletes the agent and the shared tool. For a model-swap variant deployed via `POST /agents`, delete it with `DELETE /agents/{namespace}/{name}` (the shared tool is left for other experiments).

6. End-to-end examples (validated flows)

What you need on the client side

Python version. The client side is deliberately undemanding: §6.0 needs only `curl` plus any `python3` for JSON formatting, and `benchmarking-cli` (§6.1) imports nothing beyond the standard library. Versions we have actually run:

Python	Where it was used	Result
3.12.12	recommended for a client venv — matches the workload images exactly	✓
3.12.14	the Service container itself (FROM <code>python:3.12-slim</code>)	✓
3.11.14	the repo's own venv; the declared floor (<code>requires-python = ">=3.11"</code>)	✓
3.14.3	the laptop that drove both 12-run matrices	✓

Anything ≥ 3.11 is fine. 3.12.12 is the tidiest choice because it is exactly what the exgentic agent and MCP images run, so your driver matches the workload interpreter — but nothing in the client depends on it, and the `curl` path in §6.0 is version-insensitive.

Do you have to clone the repo? For §6.0, **no** — it is `curl` and `python3` only, so a token and a Service URL are all you need. For §6.1 you need the package that provides the `benchmarking-cli` entry point, but a manual clone is still optional: the repository is public, so install it straight from git.

```
# No clone. Client only: --no-deps, because the CLI is stdlib-only and pulls in
nothing else.
uv venv --python 3.12.12 && source .venv/bin/activate
uv pip install "benchmarking-service @
git+https://github.com/webchang/benchmarking" --no-deps

# Already have a clone? Point at the repo, NOT at `.` — `-e .` looks for
pyproject.toml in the
# CURRENT directory, so running it from a scratch/testbed folder fails with
# "does not appear to be a Python project".
uv pip install -e /path/to/benchmarking --no-deps           # -e: git pull updates
the CLI in place
```

Drop `--no-deps` only if you also want to run the Service or its test suite from that venv; it adds `fastapi`, `boto3`, `pydantic` and the MCP/A2A SDKs, none of which the client needs. Keep the venv outside a cloud-synced folder (Box/Dropbox/iCloud) — a venv is thousands of small files, and sync tools are slow and occasionally destructive with it.

Where the source ends up

`<venv>/bin/benchmarking-cli` is **not** the source — it is a generated shim that does `from benchmarking_service.cli import main`. Where the real `cli.py` lives depends on how you installed, which also decides whether a `git pull` reaches you:

Install	<code>cli.py</code> lives in	Picks up upstream changes?
<code>uv pip install "...@git+https://..."</code>	<code><venv>/lib/python3.X/site-packages/benchmarking_service/</code> (a copy , built from one commit)	No — pinned. Re-run with <code>--reinstall</code>
<code>uv pip install -e /path/to/benchmarking</code>	<code>/path/to/benchmarking/src/benchmarking_service/</code> (the clone; only a finder hook is installed)	Yes, immediately
<code>uv tool install --from /path/to/benchmarking benchmarking-service</code>	<code>\$(uv tool dir)/benchmarking-service/lib/python3.X/site-packages/benchmarking_service/</code>	No — reinstall the tool
no install, <code>PYTHONPATH=<repo>/src python -m benchmarking_service.cli</code>	the clone itself	Yes, immediately

Never guess the path — ask the interpreter that is actually running it:

```
python -c "import benchmarking_service.cli as m; print(m.__file__)"
```

Note that every option installs the **whole distribution**, server modules included (`app.py`, `routes/`, `runner/`, `s3_export.py`), because they ship together. With `--no-deps` the client half still works while `import benchmarking_service.app` does not — the third-party dependencies were skipped, not the files. That the client keeps working is the point, and you can confirm it:

```
python -c "
import benchmarking_service.cli, sys
print([m for m in ('fastapi','httpx','boto3','pydantic') if m in sys.modules]
or 'no server deps loaded')"
```

6.0 Run #1 start to finish, on the ykt3 Service driving ykt2 workloads

Run #1 of the canonical matrix: gsm8k, 1 task, no gateway, no plugins. Every command below was executed exactly as written; the `run_id` and numbers are from that run. Paste the block, then the steps.

```
# ---- 0. environment (the cross-cluster instance: Service on ykt3, workloads
on ykt2/team1) ----
export SVC="https://benchmarking-rossoctl-system.apps.ykt3.hcp.res.ibm.com"
export KC="https://keycloak-keycloak.apps.ykt2.hcp.res.ibm.com" # the
instance's iss origin
export REALM=rossoctl KC_CLIENT=rossoctl KC_USER=benchmarker
export BENCH=gsm8k
export SCOPE="namespace=team1&agent=tool_calling&experiment=default"
export CURL_OPTS=-k # ykt3's route serves a self-signed cert
read -rs -p "benchmarker password: " KC_PASS; echo # never put this in a
```

file you commit

```
# ---- 1. caller token ----
export TOKEN=$(curl -s $CURL_OPTS "$KC/realms/$REALM/protocol/openid-
connect/token" \
  -d grant_type=password -d client_id="$KC_CLIENT" \
  -d username="$KC_USER" -d password="$KC_PASS" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["access_token"])')
unset KC_PASS
curl -s $CURL_OPTS "$SVC/hello" -H "Authorization: Bearer $TOKEN" | python3 -m
json.tool
# -> {"iss": ".../realms/rossoclt", "preferred_username": "benchmarker", ...}

# ---- 2. deploy the MCP tool + A2A agent (idempotent: delete first) ----
curl -s $CURL_OPTS -X DELETE "$SVC/benchmarks/$BENCH/deploy?$SCOPE" \
  -H "Authorization: Bearer $TOKEN" -o /dev/null -w 'delete -> %{http_code}\n'
# 204 or 404
sleep 10
curl -s $CURL_OPTS -X POST "$SVC/benchmarks/$BENCH/deploy" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"agent":"tool_calling","namespace":"team1","experiment":"default"}' \
  -o /dev/null -w 'deploy -> %{http_code}\n'
# 201

# ---- 3. wait until BOTH are Ready, four polls in a row ----
# One Ready reading is not enough: an agent whose MCP is not serving yet exits
and
# CrashLoopBackOffs, so readiness flaps. A run accepted mid-flap records 0
tasks.
STABLE=0; until [ "$STABLE" -ge 4 ]; do
  R=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/status?$SCOPE" -H
"Authorization: Bearer $TOKEN" \
    | python3 -c 'import sys,json; d=json.load(sys.stdin);
print(int(bool(d["tool_ready"] and d["agent_ready"])))')
  [ "$R" = 1 ] && STABLE=$((STABLE+1)) || STABLE=0
  echo "ready=$R stable=$STABLE"; sleep 10
done
# On OpenShift the Route 502s for ~10s AFTER the Service reports Ready, so gate
on the card too:
AGENT=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/status?$SCOPE" -H
"Authorization: Bearer $TOKEN" \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)
["agent_name"])')
until curl -sf -o /dev/null "https://$AGENT-
team1.apps.ykt2.hcp.res.ibm.com/.well-known/agent-card.json"; do
  echo "waiting for the agent card"; sleep 5
done; sleep 15

# ---- 4. submit the run ----
export RUN=$(curl -s $CURL_OPTS -X POST "$SVC/benchmarks/$BENCH/runs" \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"agent":"tool_calling","namespace":"team1","experiment":"default",
      "max_tasks":1,"max_parallel_sessions":1,"timeout_seconds":120}' \
  | python3 -c 'import sys,json; print(json.load(sys.stdin)["run_id"])')
echo "run_id=$RUN" # e.g. 20260902215628-
79e0713a
```

```
# ---- 5. poll to a terminal status, then print the verdict ----
while ;; do
    S=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H "Authorization:
Bearer $TOKEN" \
        | python3 -c 'import sys,json; print(json.load(sys.stdin)["status"])')
    echo "status=$S"; case "$S" in succeeded|failed|error|cancelled) break ;;
esac; sleep 10
done
# -> succeeded, summary
{"total":1,"succeeded":1,"evaluated_pass":1,"pass_rate":1.0,"wall_seconds":5.4}
# Note ~100 spans for ONE gsm8k task: ~91 are A2A/HTTP framework
internals.

# ---- 6. artifacts: list the S3 URLs, then mirror them locally ----
export PREFIX=$(curl -s $CURL_OPTS "$SVC/benchmarks/$BENCH/runs/$RUN" -H
"Authorization: Bearer $TOKEN" \
    | python3 -c 'import sys,json; print(json.load(sys.stdin)
["artifacts_prefix"])')
echo "$PREFIX"
# -> ykt3-to-ykt2/benchmark/keycloak-keycloak.apps.ykt2.hcp.res.ibm.com-
realms-rossoctl/gsm8k/20260902215628-79e0713a
mkdir -p "/tmp/benchmarking/$PREFIX" && (cd "/tmp/benchmarking/$PREFIX" && \
    for f in run.json report.ndjson report.parquet token_report.ndjson
token_report.parquet \
        span_report.ndjson span_report.parquet manifest.json; do
        curl -sS -O "https://rossoctl-benchmarking.s3.us-east-
1.amazonaws.com/$PREFIX/$f"; done && ls -la)

# ---- 7. analyse: per-task tokens, and whether the telemetry is trustworthy --
--
cd "/tmp/benchmarking/$PREFIX"
python3 - <<'EOF'
import json
rows = [json.loads(l) for l in open("report.ndjson") if l.strip()]
print("%-6s %-38s %7s %7s %5s %6s %s" % ("task", "model", "in", "out", "llm",
"tool", "passed"))
for r in sorted(rows, key=lambda x: str(x["task_id"])):
    print("%-6s %-38s %7d %7d %5s %6s %s" % (
        r["task_id"], r["model"], r["llm_input_tokens"],
r["llm_output_tokens"],
        r["llm_count"], r["tool_count"], r["evaluation_result"]))
# llm<=1 alongside tool>=2 is impossible -- every tool call needs a preceding
model turn -- so it
# means the usage-bearing span was lost and the token counts understate the
real cost.
bad = [r for r in rows if (r["llm_count"] or 0) <= 1 and (r["tool_count"] or 0)
>= 2]
print("lost token attribution:", len(bad), "of", len(rows))
EOF
# -> 0      openai/Azure/gpt-5-mini-2025-08-07      320      471      2
1 True
#      lost token attribution: 0 of 1

# Which spans did the task actually invoke? A healthy task shows TWO chat spans
-- the
```



```
# max_tokens=1 capability probe plus the real call. One means the real call's
span was lost.
python3 - <<'EOF'
import collections, json
rows = [json.loads(l) for l in open("span_report.ndjson") if l.strip()]
print("spans:", len(rows))
for kind, n in collections.Counter(r["kind"] for r in rows).most_common():
    print(" %4d %s" % (n, kind))
for r in rows:
    if r["kind"] == "chat":
        print(" chat max_tokens=%s in=%s out=%s" % (
            r["request_max_tokens"], r["input_tokens"], r["output_tokens"]))
EOF
# -> spans: 100 (other 91, phase 3, chat 2, tool 2, root 1, agent 1)
# chat max_tokens=1 in=None out=None <- the probe (rejected by a
reasoning model)
# chat max_tokens=None in=320 out=471 <- the real call

# ---- 8. tear down ----
curl -s $CURL_OPTS -X DELETE "$SVC/benchmarks/$BENCH/deploy?$SCOPE" \
-H "Authorization: Bearer $TOKEN" -o /dev/null -w 'teardown -> %
{http_code}\n' # 204
```

6.1 The same thing in one command (`benchmarking-cli`)

`benchmarking-cli` is a stdlib-only client that performs §6.0 end to end — token, pre-clean, deploy, the readiness-stability and agent-card gates, run, 424 retry, poll, artifact listing and local mirror — and exits non-zero if the run did not succeed.

```
export BM_BASE="https://benchmarking-rossoctl-system.apps.ykt3.hcp.res.ibm.com"
export BM_ISS="https://keycloak-
keycloak.apps.ykt2.hcp.res.ibm.com/realms/rossoctl"
export BM_PASSWORD_FILE="$HOME/.rossoctl-ykt3/benchmarkers.pass" # chmod 600
export BM_INSECURE=1
export BM_CARD_TEMPLATE="https://{service}-
{namespace}.apps.ykt2.hcp.res.ibm.com/.well-known/agent-card.json"

# See "What you need on the client side" above for install options; the
shortest is:
# uv pip install "benchmarking-service @
git+https://github.com/webchang/benchmarking" --no-deps
benchmarking-cli all --benchmark gsm8k --tasks 1 --timeout 120 --mirror
/tmp/benchmarking
```

Which prints, for the run above:

```
[17:56:12] agent card 200 (https://exgentic-a2a-tool-calling-gsm8k-
team1.apps.ykt2...)
[17:56:28] run_id=20260902215628-79e0713a
[17:56:38] terminal=succeeded summary={"total": 1, ..., "pass_rate": 1.0,
"wall_seconds": 5.4}
[17:56:38] artifacts_prefix=ykt3-to-ykt2/benchmarkers/keycloak-...
```



```

rossoctl/gsm8k/20260902215628-79e0713a
  json          468  https://rossoctl-benchmarking.s3.us-east-
1.amazonaws.com/.../run.json
  ...
[17:56:39] mirrored 8/8 -> /tmp/benchmarking/ykt3-to-ykt2/.../20260902215628-
79e0713a
-rw-r--r--      3002  Sep 02 17:56  manifest.json
-rw-r--r--      1035  Sep 02 17:56  report.ndjson
-rw-r--r--     11984  Sep 02 17:56  report.parquet
-rw-r--r--       468  Sep 02 17:56  run.json
-rw-r--r--     52578  Sep 02 17:56  span_report.ndjson
-rw-r--r--     10878  Sep 02 17:56  span_report.parquet
-rw-r--r--       371  Sep 02 17:56  token_report.ndjson
-rw-r--r--      4425  Sep 02 17:56  token_report.parquet
8 files, 84,741 bytes

cd /tmp/benchmarking/ykt3-to-ykt2/.../20260902215628-79e0713a

status      succeeded
pass_rate   1.0    (1/1)
wall        5s

```

The trailing `cd` is there because the mirror path is absolute and deeply nested — retyping it relative to your current directory is the easy mistake. The sizes are also the quickest check that nothing arrived truncated. Feed that directory straight into §6.0 step 7 to analyse the run.

Each step is also a subcommand, so the CLI doubles as a way to see one HTTP call at a time:

```

benchmarking-cli whoami                # GET /hello
benchmarking-cli list                  # GET /benchmarks
benchmarking-cli deploy --benchmark gsm8k
benchmarking-cli wait --benchmark gsm8k # stability + card
gates
benchmarking-cli run --benchmark gsm8k --tasks 1 # prints the run_id
benchmarking-cli poll --benchmark gsm8k --run "$RUN"
benchmarking-cli report --benchmark gsm8k --run "$RUN" # GET .../report (needs
MLflow)
benchmarking-cli artifacts --benchmark gsm8k --run "$RUN" --mirror
/tmp/benchmarking # + ls -l
benchmarking-cli teardown --benchmark gsm8k

```

No install? It runs straight from a checkout too — `python -m benchmarking_service.cli all ...` (the module imports nothing beyond the standard library, so the server's dependencies are not needed to drive the API).

Useful options: `--parallel` (`max_parallel_sessions`), `--task-timeout`, `--model` for a deploy-time swap, `--preset` `auth-only|ibac-only|full` and repeatable `--plugin` `ibac:observe` for AuthBridge, `--no-deploy` to reuse a deployment, `--teardown` to clean up afterwards. Switching to KinD needs only three lines:

```
export BM_BASE="http://benchmarking.localtest.me:8080"
export BM_ISS="http://keycloak.localtest.me:8080/realms/rossoctl"
export BM_PASSWORD_FILE="$HOME/.rossoctl-kind/benchmarkers.pass"; unset
BM_INSECURE BM_CARD_TEMPLATE
```

For the full 12-run matrix use [reference/run-12.py](#) instead — see §6.3. It implements the same gates independently; consolidating both behind `benchmarking_service.cli` is a known follow-up.

6.2 Other validated flows

These map to the canonical parameterized runs and were validated on `ykt3` with S3 export.

```
# Run #2 – gsm8k, 10 tasks (validated: 9/10 pass)
curl -s -X POST "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 10, "max_parallel_sessions": 1, "timeout_seconds": 600}'

# Run #3 – gsm8k, 50 tasks, 4-way concurrency (validated: 42/50, ~190s)
curl -s -X POST "$SVC/benchmarks/gsm8k/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 50, "max_parallel_sessions": 4, "timeout_seconds": 900}'

# Run #9 – tau2, 10 tasks (multi-turn; deploy tau2 first)
curl -s -X POST "$SVC/benchmarks/tau2/deploy" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" -d '{"namespace": "team1"}'
curl -s -X POST "$SVC/benchmarks/tau2/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 10, "max_parallel_sessions": 1, "timeout_seconds": 1200}'

# Run #10 – tau2, 20 tasks, 4-way concurrency (validated: 17/20, ~915s – needs
# raised timeout)
curl -s -X POST "$SVC/benchmarks/tau2/runs" -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"max_tasks": 20, "max_parallel_sessions": 4, "timeout_seconds": 1800}'
```

6.3 The whole 12-run matrix in one command ([reference/run-12.py](#))

Same idea as §6.1, one level up: instead of a single benchmark run, this drives all **12 canonical parameterized runs** — gsm8k at three sizes, a model swap, the four AuthBridge presets, tau2 at two sizes, appworld at two — deploying each leg fresh, mirroring every artifact, and writing one state file the report generators read.

Unlike `benchmarking-cli`, this one **is** repo-only: it is a matrix driver tied to `run12_specs.json`, not something you install. Clone first.

```
git clone https://github.com/webchang/benchmarking && cd benchmarking

# Same five variables as §6.1 – plus a label, which names the state file and
```

```

the reports.
export BM_BASE=https://benchmarking-rossoctl-system.apps.ykt3.hcp.res.ibm.com
export BM_ISS=https://keycloak-
keycloak.apps.ykt2.hcp.res.ibm.com/realms/rossoctl
export BM_PASSWORD_FILE="$HOME/.rossoctl-ykt3/benchmarkers.pass"
export BM_INSECURE=1
export BM_CARD_TEMPLATE="https://{service}-
{namespace}.apps.ykt2.hcp.res.ibm.com/.well-known/agent-card.json"
export BM_LABEL=ocp-v124

python3 reference/run-12.py                # ~1h55m; log it: ... 2>&1 | tee
/tmp/run12-ocp.log

```

It ends with a per-leg summary and the state file path:

```

[21:03:57] === done in 6884s; 12 runs -> /tmp/benchmarking/run12-ocp-v124.json
[21:03:57]  #1  gsm8k      succeeded      pass=1.0   tasks=1
[21:03:57]  #2  gsm8k      succeeded      pass=1.0   tasks=10
...
[21:03:57]  #12 appworld  succeeded      pass=0.0   tasks=20

```

Then turn that state file into the three documents (all numbers derived from the mirrored artifacts — nothing transcribed):

```

# 1. the 12-run report. <date> is the date the RUNS executed, from the modal
run_id prefix.
python3 reference/gen-12run-report.py /tmp/benchmarking/run12-ocp-v124.json
v1.24 \
  "OpenShift - Service on ykt3, workloads on ykt2" results/12run-report-v1.24-
20260901-ocp.md

# 2. OCP vs KinD, once both matrices exist
python3 reference/gen-12run-comparison.py \
  /tmp/benchmarking/run12-ocp-v124.json OCP \
  /tmp/benchmarking/run12-kind-v124.json KinD v1.24 \
  results/12run-comparison-v1.24-20260902.md

# 3. AuthBridge plugin overhead. The judge log is what tells it how many tool
calls were actually
#   authorized - without it, per-preset medians mix judged and unjudged calls
and mislead.
oc -n rossoctl-system logs deploy/ibac-judge --since=6h --timestamps \
  | grep -vi healthz | awk '{print $1}' > /tmp/judge-ocp.ts
PEER_JSON=/tmp/benchmarking/run12-kind-v124.json PEER_LABEL="single-node KinD"
\
PEER_JUDGE_TS=/tmp/judge-kind.ts \
python3 reference/gen-plugin-overhead.py /tmp/benchmarking/run12-ocp-v124.json
v1.24 \
  "OpenShift - Service on ykt3, workloads on ykt2" \
  results/12run-plugin-overhead-v1.24-20260901-ocp.md /tmp/judge-ocp.ts

```

Switching to KinD is the same three-line change as §6.1 plus a new label:

```
export BM_BASE=http://benchmarking.localtest.me:8080
export BM_ISS=http://keycloak.localtest.me:8080/realms/rossoctl
export BM_PASSWORD_FILE="$HOME/.rossoctl-kind/benchmarkers.pass"
unset BM_INSECURE BM_CARD_TEMPLATE
export BM_LABEL=kind-v124
python3 reference/run-12.py # ~1h55m on a single node too
```

Run the two platforms sequentially, not in parallel — they share the external LLM gateway, so overlapping them makes the wall-time comparison meaningless.

Other knobs, all optional:

Variable	Default	Use
BM_ONLY	all 12	comma-separated leg numbers, e.g. BM_ONLY=7 to re-run one leg, BM_ONLY=1,2,3 for a smoke pass
BM_STABLE_POLLS	4	consecutive ready polls required before a run is submitted
BM_SETTLE_PLAIN / BM_SETTLE_SIDECAR	15 / 45	post-ready settle seconds; sidecar deploys need longer
BM_USER / BM_CLIENT	benchmarker / rossoctl	non-default realm identity
BM_PASSWORD	—	the password inline, instead of BM_PASSWORD_FILE . Prefer the file (chmod 600): an exported variable leaks into env , process listings and shell history

BM_ONLY is how you repair a matrix without redoing it: a leg that failed to deploy can be re-run on its own and merged into the state file, which is exactly what happened to leg #7 of the v1.24 OCP matrix (a transient **DELETE -> 500** left a stale tool behind and the deploy hit **409**).

7. Known limits & error codes

Situation	Code	What to do
Missing/invalid bearer	401	fetch a fresh token (§2)
iss not in any instance file	403	add/repair the instance config (§3.1)
/config as non-benchmarker	403	authenticate as the benchmarker user
Setting a workload cred via /config	422	provision it as a cluster Secret instead (§3.3)
plugin_preset/plugins/on_error without authbridge_enabled=true	422	set authbridge_enabled=true so the sidecar is injected for the pipeline to take effect
plugin_config_file on deploy	422	local-path input with no HTTP analog; use plugin_preset/plugins/on_error instead

Situation	Code	What to do
Run before deploy	409	POST .../deploy first
Run while not Ready	424	provision the named Secret(s), wait for Ready (§5.2)
Report with MLflow unconfigured	409	set MLflow read creds via PUT /config (§3.4)
Upstream Rossoctl/MLflow failure	502	transient upstream issue; retry

AuthBridge plugin presets (runs 4–8 of the ibac comparison): layer-3 plugin composition and `on_error` selection require overlaying the per-agent `authbridge-config-<agent>` ConfigMap via `kubectl` — which the HTTP-only Service cannot do — so those are honestly rejected with 422. The one enactable knob is `authbridge_enabled=true` (injects the sidecar with the cluster-default pipeline).

appworld: registers and deploys correctly via the Service, but its MCP image hangs on its own per-task Venv-service health timeout (120s), independent of the Service. Treat appworld as externally blocked until that image is fixed.

8. Extending the catalog (adding or changing a benchmark)

The benchmark catalog is **code, not runtime data** — a static `BENCHMARKS` dict in `src/benchmarking_service/benchmarks/registry.py`. There is no database, config file, or admin API behind it, and the HTTP surface over the catalog is **read-only** (`GET /benchmarks`, `GET /benchmarks/{name}`). Adding or changing a benchmark is therefore a **source change + tests + image rebuild + redeploy**, deliberately: the definition pins container images, dataset env, and resource limits that must move in lockstep with the workload images, so every change is a reviewable, git-tracked, image-versioned artifact rather than mutable state that could drift per instance.

See the co-located contributor note `src/benchmarking_service/benchmarks/README.md` for the field-by-field reference and the per-benchmark env gotchas.

8.1 What a definition holds

Each entry is a `BenchmarkDefinition`: `name`, `mcp_image` (+ `mcp_image_tag/mcp_port/mcp_path`), `tool_env`, `tool_resources`, `default_model`, `user_simulator` (multi-turn flag), and an `agents` map of `BenchmarkAgentSpec` (per-agent `container_image` + `extra_env` + `resources`). Secret references are declared with the `_secret_env(env, secret, key)` helper; `required_secrets()` derives the actionable 424 precheck message from them automatically.

8.2 Add a new benchmark

1. Add a `BenchmarkDefinition` entry to `BENCHMARKS` in `registry.py`:

- `mcp_image` (+ `tag/port/path`) — the MCP tool image.
- `tool_env` — `BENCHMARK_NAME`, the LiteLLM base URL, and any secret refs via `_secret_env`.
- `agents={...}` — one `BenchmarkAgentSpec` per flavor. The `tool_calling` A2A image is shared across all current benchmarks; usually you reuse it and only the name gets a `-<benchmark>` suffix.
- `user_simulator=True` **only** for multi-turn benchmarks — this is what makes `build_tool_request` inject `EXGENIC_SET_BENCHMARK_USER_SIMULATOR_MODEL` into the MCP pod so the server-side simulator shares the run's model.

- Any benchmark-quirk env — mind the documented gotchas: gsm8k needs `EXGENTIC_SET_BENCHMARK_RUNNER=direct`; tau2 needs `EXGENTIC_SET_BENCHMARK_ACTION_TIMEOUT=1000`; appworld **rejects** that same action-timeout override and crashes at startup if it's present.
- 2. **Add tests** in `tests/test_benchmarks.py` — assert `build_tool_request` / `build_agent_request` emit the expected images, env, secrets, and resources, mirroring the existing patterns.
- 3. **Rebuild + bump the image tag**, redeploy the Service. The new benchmark then appears in `GET /benchmarks` and is deployable/runnable at `/benchmarks/<newname>/...` with **no client change** — the request bodies are benchmark-agnostic (see §5).

8.3 Change an existing benchmark

Same mechanism — edit the entry (bump `mcp_image_tag`, change `default_model`, add an env var, adjust `tool_resources`), update tests, rebuild, redeploy. Because `required_secrets()` is derived from `tool_env` + the chosen agent's `extra_env`, changing a secret reference automatically updates the 424 "this benchmark requires secret(s): ..." precheck message — no separate wiring.

8.4 What stays runtime-mutable (for contrast)

Per-instance state is *not* in the catalog and does not require a rebuild: instance config (files under `settings.instances_dir`, keyed by `iss`) and the benchmarker-only `PUT /config` overrides (MLflow read creds + S3). The benchmark catalog is intentionally the immutable, version-pinned part.